

Unneeded faces inside repeater models

MC-177452 Textures and models

Status	Resolved / Fixed
Affected	1.15.2, 20w14a, 20w18a, 20w21a, 1.16 Pre-release 2, 1.16 Pre-release 3 (+28 more)
Fixed in	24w33a
Reported	2020-04-04

STEPS TO REPRODUCE

1. Launch Minecraft.
2. Create or enter a Creative world with default settings.
3. Place an unpowered redstone repeater.
4. Enter Spectator mode.
5. Fly into the repeater until the bottoms of its redstone torches are in view.
6. Observe that the unpowered torches have visible bottom faces despite those faces only being viewable by clipping into the repeater.

OBSERVED BEHAVIOUR

Actual behaviour

The repeater block models include unnecessary bottom faces for geometry that is never exposed during normal gameplay:

- The unlit redstone torch elements include faces on their undersides.
- The lock elements in the locked-repeater models also include bottom faces.

These faces are internal to the repeater and can only be seen by moving the camera inside the block, such as in Spectator mode. They therefore provide no visible surface in normal use, but are still present in the model JSON and are processed by the renderer when the geometry is clipped into. This adds redundant geometry to the affected repeater models.

EXPECTED BEHAVIOUR

The repeater model should omit the bottom faces of its unlit redstone torches and the bottom faces of its lock components, since these surfaces cannot be seen during normal gameplay. Even when viewed by entering the repeater in Spectator mode, no unnecessary bottom geometry should be rendered, keeping the model limited to faces that can be visible.

ENVIRONMENT

Minecraft Java Edition Versions: 1.15.2, 20w14a, 20w18a, 20w21a, 1.16 Pre-release 2, 1.16 Pre-release 3, 1.16.1, 20w27a, 20w28a, 20w29a, 20w30a, 1.16.2 Pre-release 1, 20w49a, 20w51a, 21w03a, 1.16.5, 21w05b, 21w06a, 21w08b, 21w15a, 21w16a, 21w20a, 1.17 Pre-release 1, 1.17, 1.17.1, 1.18, 1.18.1, 1.18.2, 1.19, 1.19.2, 1.20 Pre-release 1, 1.20.1, 23w43a, 1.21 (includes snapshots)

ORIGINAL DESCRIPTION

The resource pack attached to
MC-214686

Resolved

offers a complete fix for this issue. Mojang have my full permission to (and are encouraged to) replace the affected vanilla model files with the contents of the resource pack.

The bug

Redstone repeater models define faces for the bottoms of unlit redstone torches, as well as the bottom of locks for locked repeaters, despite these never being visible without clipping into the repeater itself. Powered torches correctly omit such faces due to not using a single cuboid.

How to reproduce

- Place down an unpowered redstone repeater
- Enter Spectator mode
- Fly into the repeater such that the bottoms of the torches are in view

Expected behaviour

The unpowered torches would not have any bottom faces defined since such faces are never possible to see in normal gameplay.

Actual behaviour

These bottom faces are defined anyway, bloating the model file and forcing the game to render useless faces.

How to fix

The resource pack attached to

MC-214686

Resolved

fixes this issue completely.

MANUAL RESULT

Version used	
Reproduced?	YES / NO / PARTIAL
Crash file	
Notes	
Tested by / date	

Live issue: <https://bugs.mojang.com/browse/MC/issues/MC-177452>